

BORONG ZHANG

bzhang388@wisc.edu | Madison WI 53706
borongzhang.com | github.com/borongzhang

EDUCATION

University of Wisconsin–Madison

Ph.D. Candidate in Mathematics, Advisors: Prof. Qin Li & Leonardo Andrés Zepeda Núñez Madison, WI 09/2020 – 06/2026

- **Honors:** Mathematics Department Ascending Scholar Fellowship

University of California, Berkeley

Bachelor of Arts in Applied Mathematics & Computer Science Berkeley, CA 08/2016 – 06/2020

- **Honors:** High Honors in Applied Mathematics; Distinction in General Scholarship; Dean's Honors Lists

PUBLICATIONS

Peer-reviewed publications

Zhang, B., Guerra, M., Li, Q., & Zepeda-Núñez, L. (2025). Back-Projection Diffusion: Solving the wideband inverse scattering problem with diffusion models. *Computer Methods in Applied Mechanics and Engineering*, 443, 118036. <https://doi.org/10.1016/j.cma.2025.118036>

Zhang, B., Zepeda-Nunez, L., & Li, Q. (2024). Solving the wide-band inverse scattering problem via equivariant neural networks. *Journal of Computational and Applied Mathematics*, 451, 116050–116050. <https://doi.org/10.1016/j.cam.2024.116050>

Huang, E. G., Wang, R.-Y., Xie, L., Chang, P., Yao, G., **Zhang, B.**, Ham, D. W., Lin, Y., Blakely, E. A., & Sachs, R. K. (2020). Simulating galactic cosmic ray effects: Synergy modeling of murine tumor prevalence after exposure to two one-ion beams in rapid sequence. *Life Sciences in Space Research*, 25, 107–118. <https://doi.org/10.1016/j.lssr.2020.01.001>

Preprints

Zhang, B., Deng, J., Jiang, Y. & Di, Z. W. (2025). Stochastic Multigrid Method for Blind Ptychographic Phase Retrieval. *ArXiv.org*. <https://arxiv.org/abs/2511.01793>

Zhang, B., Li, Q., & Di, Z. W. (2025). MAGPIE: Multilevel-Adaptive-Guided Solver for Ptychographic Phase Retrieval. *ArXiv.org*. <https://arxiv.org/abs/2504.10118>. *Conditionally accepted at SIAM Journal on Scientific Computing, pending minor revision.*

SELECTED TALKS

Efficient Symmetry-Driven Diffusion Models for Wideband Inverse Scattering 10/2025

- SIAM Central States Section Annual Meeting Fayetteville, AR

Efficient Symmetry-Driven Diffusion Models for Wideband Inverse Scattering (Lightning Talk) 06/2025

- The Institute for Mathematical and Statistical Innovation Chicago, IL

Efficient Symmetry-Driven Diffusion Models for Wideband Inverse Scattering 05/2025

- SIAM Conference on Applications of Dynamical Systems Denver, CO

Multigrid-based Stochastic Minimization for Ptychographic Phase Retrieval 04/2025

- Copper Mountain Conference On Iterative and Multigrid Methods Frisco, CO

Solving the Inverse Scattering Problem: Leveraging Symmetries for Machine Learning 11/2024

- SIAM Student Chapter Seminar, University of Wisconsin-Madison Madison, WI

PROFESSIONAL SERVICE

Reviewer, International Conference on Machine Learning 03/2026

Co-organizer (upcoming), Minisymposium 11/2026

- SIAM Conference on Mathematics of Data Science Salt Lake City, UT

CONFERENCES, WORKSHOPS & SUMMER SCHOOLS

SIAM Central States Section Annual Meeting 10/2025

- University of Arkansas Fayetteville, AR

Statistical and Computational Challenges in SciML 06/2025

- The Institute for Mathematical and Statistical Innovation Chicago, IL

SIAM Conference on Applications of Dynamical Systems	05/2025
• Sheraton Denver Downtown Hotel	Denver, CO
Copper Mountain Conference On Iterative and Multigrid Methods	04/2025
• Copper Mountain	Frisco, CO
Atlanta SIAM Student Conference	03/2025
• Georgia Institute of Technology	Atlanta, GA
Data-driven PDE-based Inverse Problem, in Theory and Practice	08/2024
• University of Wisconsin-Madison	Madison, WI
Junior Researcher Meeting, on Forward and Inverse Kinetic theory	09/2022
• University of Wisconsin-Madison	Madison, WI

RESEARCH EXPERIENCE & INTERNSHIPS

Givens Associate, MCS Division, Argonne National Laboratory	05/2025 - 08/2025
Supervisor: Dr. Zichao (Wendy) Di	Lemont, IL
- Proposed a stochastic multigrid method for blind ptychographic phase retrieval that employs a geometric-mean, phase-aligned joint object–probe update, with convergence guarantees. - Designed, implemented, and benchmarked the algorithm in Python (NumPy/PyTorch).	

Research Aide, MCS Division, Argonne National Laboratory	06/2024 - 09/2024
Supervisor: Dr. Zichao (Wendy) Di	Lemont, IL
- Proposed a stochastic multigrid, majorization–minimization framework for ptychographic phase retrieval. - Delivered a comprehensive theoretical analysis of convergence. - Designed, implemented, and benchmarked the algorithm in Python.	

TEACHING EXPERIENCE

Math 718: Randomized Linear Algebra and Applications	Spring 2026
Math 340: Elementary Matrix and Linear Algebra, TA	Fall 2025
Math 211: Calculus, Head TA	Spring 2025
Math 112: College Algebra, Instructor	Fall 2023-Fall 2024
Math 234: Calculus - Functions of Several Variables, TA	Spring 2023
Math 211: Calculus, TA	Fall 2022
Math 521: Analysis I, TA	Summer 2022
Math 222: Calculus and Analytic Geometry II, TA	Spring 2021
Math 221: Calculus and Analytic Geometry I, TA	Fall 2020

ORGANIZATIONS & OUTREACH

Graduate Peer Mentor	Fall 2025
Directed Reading	Summer 2025-Fall 2025
• Topic: Stochastic Multigrid Methods for Ptychographic Phase Retrieval	
Directed Reading Program, Mentor	Spring 2025
• Topic: Stochastic Differential Equations: Score-Based Diffusion Models	
Directed Reading Program, Mentor	Fall 2024
• Topic: Solving the Inverse Scattering Problem: Classical Methods and Machine Learning	

SKILLS & INTERESTS

Programming Languages: Python, Java, MATLAB, Julia, R, C
Libraries, APIs, and Technologies: Git, Jupyter, SciPy, NumPy, JAX, Flax, TensorFlow, PyTorch